WEEK 11 PROJECT REPORT

Automobile Data Cleaning & Transformation

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1. Introduction

This project shows how we cleaned and changed the Automobile data using Microsoft Power BI. The main goal for Week 11 is to take the raw car data and create new columns.

We created new columns to find the price per horsepower, power-to-weight ratio, and fuel rating groups. We also built simple formulas to find average car values. Finally, we checked all our steps and loaded the clean data into Power BI.

2. Objectives

• Load the Automobile data (Automobile_data.csv) into Power BI Desktop.

• Check the data to make sure all rows and columns look right.

• Replace question marks (?) with empty spaces (null).

• Fix data types so text is text and numbers are numbers (like price, horsepower, and peak-rpm).

• Create a Price per Horsepower column to see cost efficiency.

• Create a Power-to-Weight Ratio column to check engine power against car weight.

• Create an MPG Rating Category column to group cars by fuel mileage.

• Create a separate _Measures table to hold average formulas.

• Build a clean automobile data model ready for grouping by body-style, drive-wheels, and fuel-system.

3. Tools Used

• Microsoft Power BI Desktop: The main software used to load data and build the model.

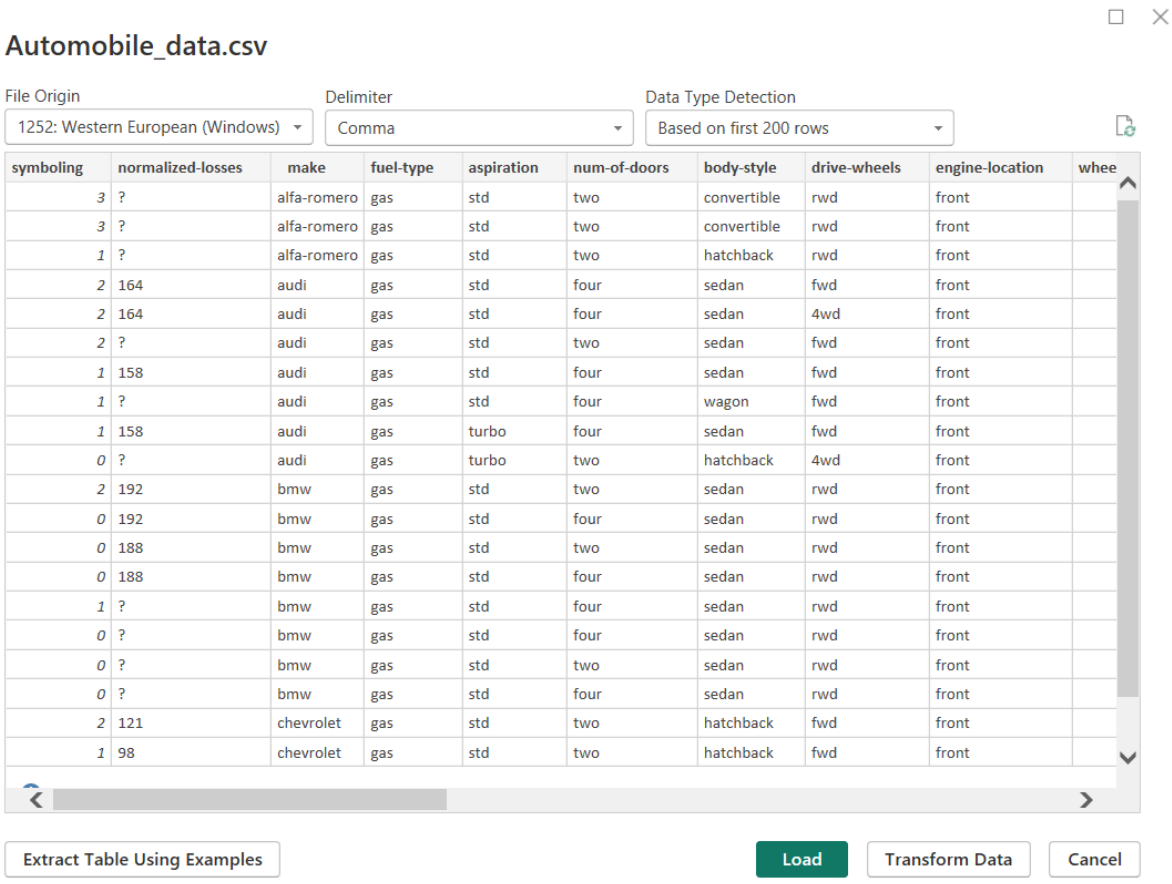
• Power Query Editor: The tool inside Power BI used to clean data and add columns.

• DAX: The formula language used to create average measures.

4. Step-by-Step Work

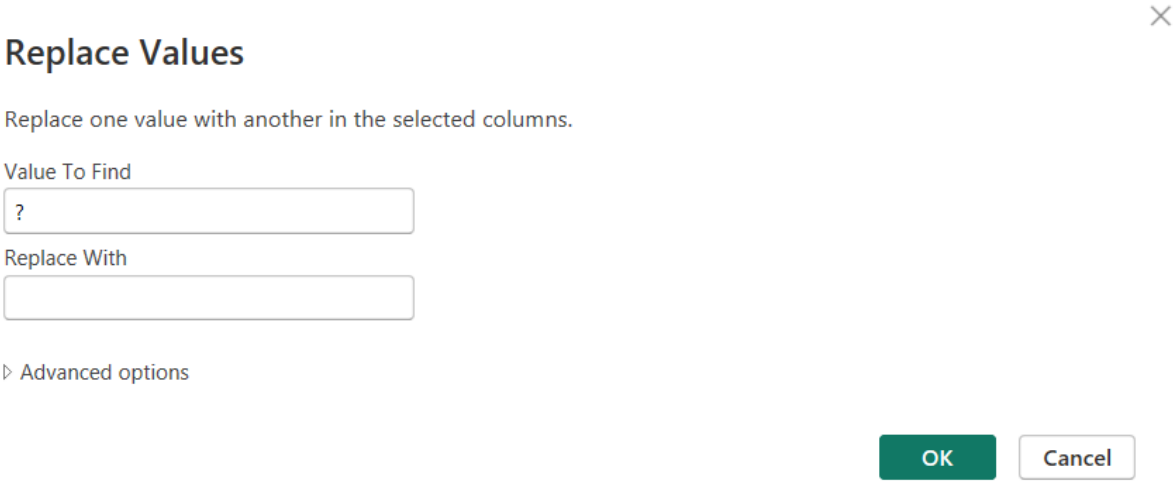
Step 1 – Loading and Cleaning the Data

• Load the Automobile CSV file into Power BI.

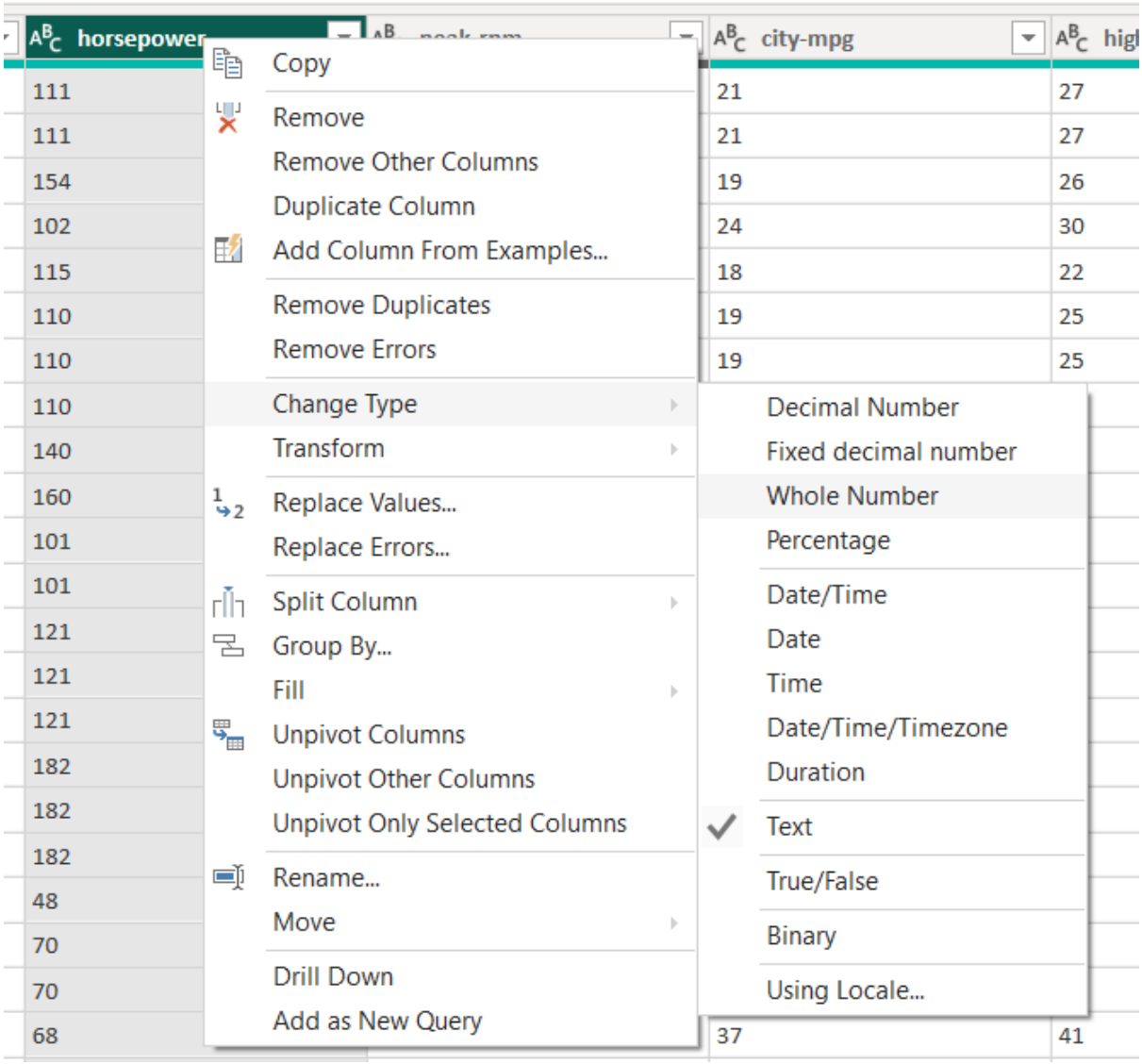


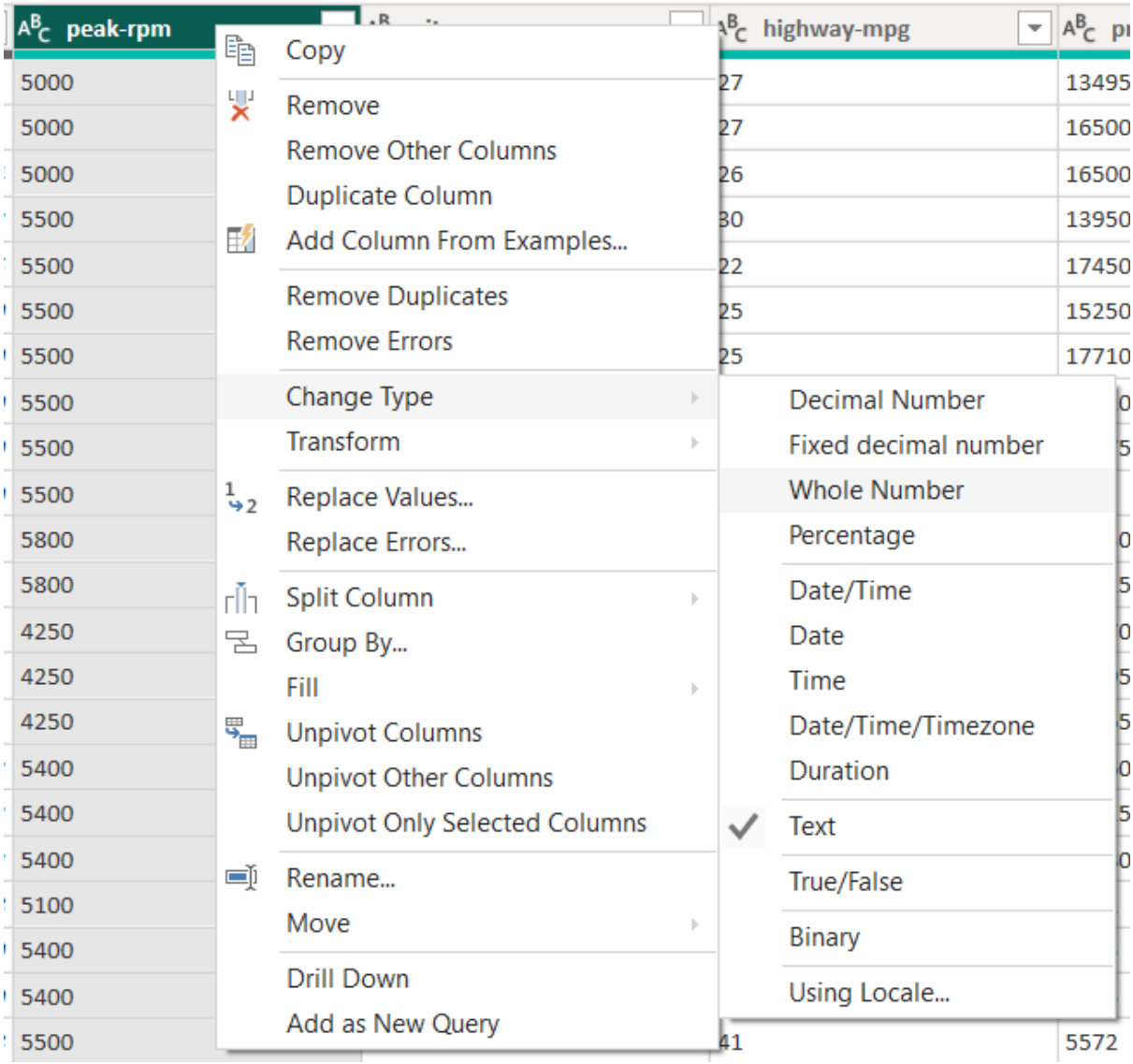
symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	whee
3	?	alfa-romero	gas	std	two	convertible	rwd	front	
3	?	alfa-romero	gas	std	two	convertible	rwd	front	
1	?	alfa-romero	gas	std	two	hatchback	rwd	front	
2	164	audi	gas	std	four	sedan	fwd	front	
2	164	audi	gas	std	four	sedan	4wd	front	
2	?	audi	gas	std	two	sedan	fwd	front	
1	158	audi	gas	std	four	sedan	fwd	front	
1	?	audi	gas	std	four	wagon	fwd	front	
1	158	audi	gas	turbo	four	sedan	fwd	front	
0	?	audi	gas	turbo	two	hatchback	4wd	front	
2	192	bmw	gas	std	two	sedan	rwd	front	
0	192	bmw	gas	std	four	sedan	rwd	front	
0	188	bmw	gas	std	two	sedan	rwd	front	
0	188	bmw	gas	std	four	sedan	rwd	front	
1	?	bmw	gas	std	four	sedan	rwd	front	
0	?	bmw	gas	std	four	sedan	rwd	front	
0	?	bmw	gas	std	two	sedan	rwd	front	
0	?	bmw	gas	std	four	sedan	rwd	front	
2	121	chevrolet	gas	std	two	hatchback	fwd	front	
1	98	chevrolet	gas	std	two	hatchback	fwd	front	

• Replace Values to find all question marks (?) and replace them with empty spaces.



• Change Type to turn horsepower, peak-rpm, and price into proper numbers.

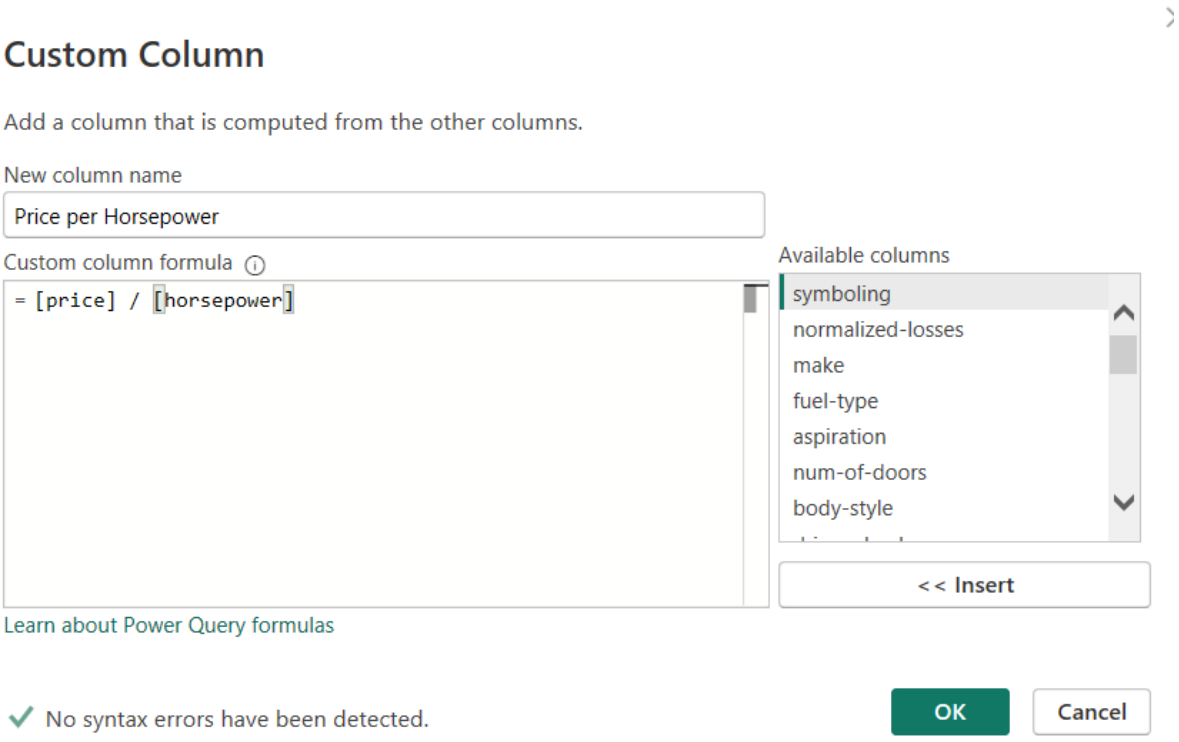

horsepower
city-mpg
Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Split Column
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move
Drill Down
Add as New Query
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...



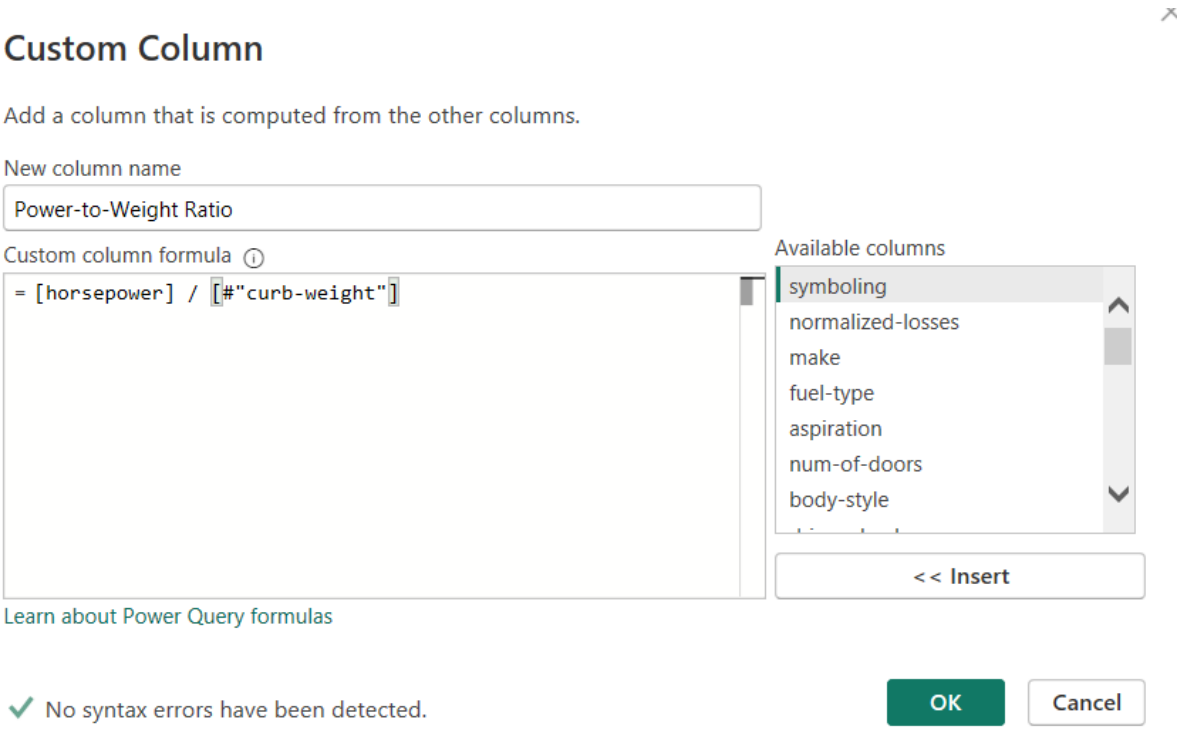
Step 2 – Creating Custom Columns

Add Column tab and click Custom Column to make two new columns:

• Price per Horsepower: We divided price by horsepower.

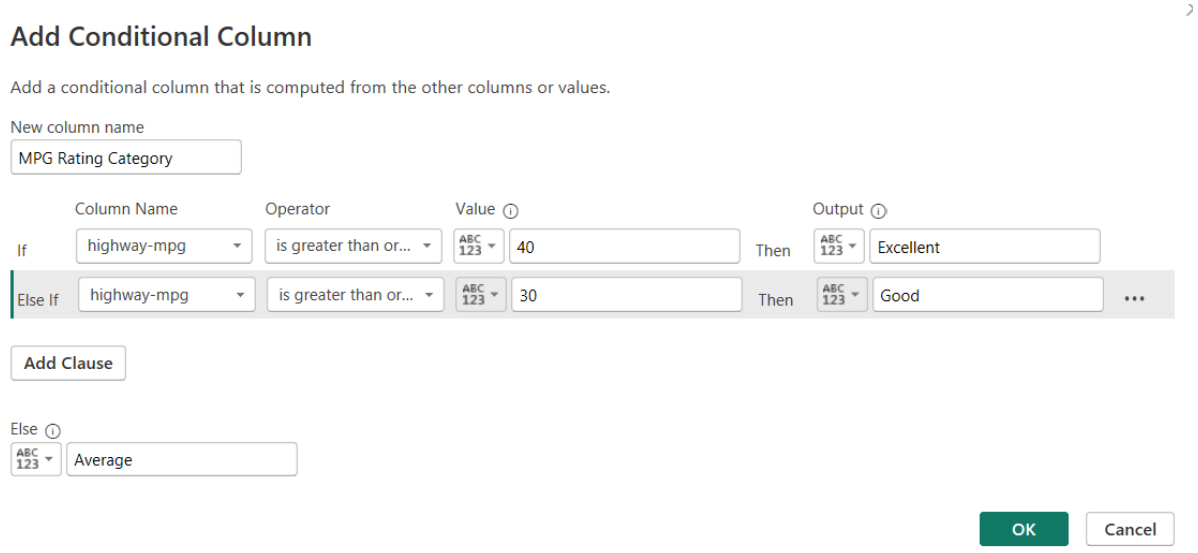


- Power-to-Weight Ratio: We divided horsepower by curb-weight.



Step 3 – Creating Conditional Groups

- Click Conditional Column to create the MPG Rating Category based on highway mileage:
    - 40 or more → Excellent
    - 30 or more → Good
    - Everything else → Average



Step 4 – Creating Separate DAX Measures

- Click Enter Data to make an empty table named _Measures to keep our formulas.

1. Avg Price = AVERAGE(Automobile_data[price])
2. Avg Horsepower = AVERAGE(Automobile_data[horsepower])
3. Avg Engine Size = AVERAGE(Automobile_data[engine-size])
4. Avg Highway MPG = AVERAGE(Automobile_data[highway-mpg])

5. Conclusion

In Week 11, we cleaned and improved the Automobile data using Power Query and DAX. By adding new columns, we can easily see the cost of horsepower, engine performance, and fuel ratings. Putting the average formulas into a separate table.